

Institute of Actuaries of India

Subject CT7 – Business Economics

May 2013 Examinations

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solutions 1 to 30 :

- 1) D
- 2) B
- 3) C
- 4) C
- 5) D
- 6) C
- 7) D
- 8) D
- 9) B
- 10) D
- 11) C
- 12) C
- 13) D
- 14) B
- 15) C
- 16) D
- 17) C
- 18) D
- 19) D
- 20) D
- 21) D
- 22) C
- 23) D
- 24) B
- 25) D
- 26) B
- 27) No option is incorrect; so award full marks to every candidate
- 28) C
- 29) D
- 30) B

[Q.No. 1 to 30 =45 Marks]

Solution 31:

Globalization refers to the process of developing increasing political, cultural and economic ties between people all around the world. Recently it has been more rapid due to dramatic improvements in transport and communication links.

The potential for the globalisation of an industry is driven by four main factors:

- 1. *market drivers*
- 2. *cost drivers*
- 3. *government drivers*
- 4. *competitive drivers*

Other drivers include improved communications (*eg* the internet) and the globalization of financial markets.

Supporters of globalisation argue that the benefits include:

- increased opportunities for specialization and the exploitation of economies of scale, leading to increased production of goods and services and higher profits
- the faster diffusion of new technology
- greater competition, lower prices and a wider range of goods for consumers
- increased investment in developing countries, leading to growth and improvements in living standards
- closer political ties and hence greater political stability
- greater cultural exchange to the benefit of people in different countries.

Critics of globalisation argue that it:

- contributes to growing inequalities between countries
- makes poor countries even poorer as multinationals exploit their dominant position in foreign markets
- contributes to environmental problems, as rapid growth and increased transportation cause pollution and the depletion of natural resources
- leads to political, economic and cultural domination by large, multinational brands.

[Total-7]

Solution 32:

a. In Actuaría, 3 more units of Bikes can be produced by giving up 5 units of Cars. The opportunity cost ratio is $3B \equiv 5C$, or $.6B \equiv 1C$.

b. In Economía, 10 additional units of Bikes can be produced by giving up 8 units of Cars. The opportunity cost ratio is $10B \equiv 8C$, or $1.25B \equiv 1C$.

c. Actuaría has a lower opportunity cost of Cars while Economía has a lower opportunity cost of Bikes. Actuaría should specialize in Cars and Economía in Bikes.

d. Prior to specialization, the two countries' combined cars production was 18 (10 in Actuaría and 8 in Economía). Total bikes production was 36 (6 + 30). Following specialization, Economía will produce 40 units of Bikes (a gain of 4 units) and Actuaría 20 units of Cars (a gain of 2.)

e. The terms of trade can vary anywhere between the pretrade opportunity cost ratios. That is, anywhere between 1 unit of Cars for 0.6 units of Bikes and 1 unit of Cars for 1.25 units of Bikes.

f. Following trade, Actuaría would have 10 units of Cars (production of 20 minus exports of 10) and 10 units of Bikes. Compared to its pre-trade choice at C, it has gained 4 units of Bikes. Economía would obtain 10 units of Cars and retain 30 units of Bikes (production of 40 minus its exports of 10.) This is a gain of 2 units of Cars compared to its pre-trade position.

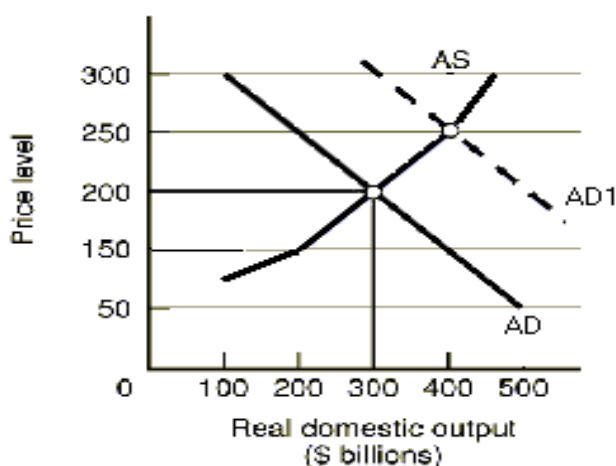
[Total-6]

Solution 33:

- a) Using the expenditure approach, GDP is the sum of $C + I_g + G + X_n$. In this example, $GDP = 525 + 110 + 72 - 15 = 692$. The income approach to GDP combines compensation of employees (462), proprietors' income (59), interest (29), rents (26), corporate profits (75), taxes on production and imports (22), and consumption of fixed capital (31), less net foreign factor income (12). These also sum to 692.
- b) NDP, or net domestic product, is equal to GDP less depreciation. $NDP = 692 - 31 = 661$.
- c) $NI = NDP + \text{net foreign factor income}$. $NI = 661 + 12 = 673$. Alternatively, NI equals the sum of compensation of employees, proprietors' income, interest, rents, corporate profits, and taxes on production and imports. $NI = 462 + 59 + 29 + 26 + 75 + 22 = 673$.
- d) $PI = NI$ less taxes on production and imports, Social Security contributions, corporate income taxes and undistributed corporate profits, plus transfer payments. $PI = 673 - 22 - 39 - 28 - 17 + 33 = 600$.
- e) $DI = PI$ minus personal taxes. $DI = 600 - 71 = 529$.

[Total-6]**Solution 34:**

- (c) See the graph. Equilibrium price level = 200. Equilibrium real output = 300 billion. No, the full-capacity level of GDP is more likely at 400 billion, where the AS curve starts to become steeper.
- (b) At a price level of 150, real GDP supplied is a maximum of 200 billion, less than the real GDP demanded of 400 billion. The shortage of real output will drive the price level up. At a price level of 250, real GDP supplied is 400 billion, which is more than the real GDP demanded of 200 billion. The surplus of real output will drive down the price level. Equilibrium occurs at the price level at which AS and AD intersect.
- © See the graph. Increase in number of consumers, investment, government, or net export spending might shift the AD curve rightward. New equilibrium price level = 250. New equilibrium GDP = 400 billion.



[Total-6]

Solution 35:**Validation:**

Banks make profits by lending at a higher rate of interest than that paid to depositors. Ideally, banks would like to “borrow short-term” (at low rates of interest) and “lend long-term” (at higher rates of interest).

Maturity gap is the difference in the average maturity of loans and deposits, the larger the maturity gap, the greater the profitability.

Explanation:

However, banks have to be able to meet their customers’ demands for cash. The liquidity of an asset is the ease with which it can be converted into cash without loss. Cash is perfectly liquid and short-term loans are more liquid than long-term loans and investments. Therefore, a larger maturity gap reduces liquidity. The liquidity ratio is the proportion of a bank’s assets held in liquid form (cash, balances at the central bank, short-term loans, and government bonds with a year to maturity).

Banks therefore have to balance their desire for profitability (requiring a low liquidity ratio) with the need to avoid a financial panic (requiring a high liquidity ratio).

This shows that banks must not only consider the *length* of the loans; they must also consider the *riskiness* of the loans they make, *ie* the likelihood of default, and must be able to cover any losses arising from such defaults.

For example, short-term loans to other banks are regarded as liquid assets, but, if the borrowing banks are in danger of insolvency, such lending is very risky.

[Total-7]

Solution 36:

- a) The balance on goods is equal to net exports (exports minus imports), or $100 - 130 = -\$30$ billion.

The balance on services is equal to net services exports, or $50 - 45 = \$5$ billion

The balance on goods and services is the sum of the balance on goods and the balance on services, or $-30 + 5 = -\$25$ billion

The balance on current account is the balance on goods and services (-25) plus net investment income (-5) and net transfers (+15), or $-\$15$ billion

- b) The balance on capital and financial account is sum of the balance on capital account (-5) and the balance on financial account (+20), or equal to $+\$15$ billion.
- c) The balance of payments sums the balance on current account and the balance on capital and financial account. For Monteria, the balance of payments shows a zero balance: the \$15 billion deficit on current account exactly offsets the \$15 billion surplus on the capital and financial account.

[Total-3]

Solution 37:

Jaspreet has two options to work on: 1. Work for university maintenance crew, or 2. Run pizza stand at the college canteen building. These two alternatives are mutually exclusive and hence presents opportunity cost situation for Jaspreet. Therefore, Jaspreet needs to compare opportunity cost in terms of benefits foregone if he run pizza stand at the college canteen.

If Jaspreet runs pizza stand:

$TR = 100 \times 60 = 6000/-$ spending a total of $10 \times 6 = 60$ hours of time

$TC = \text{Fixed Cost} + \text{Variable Cost} = 1500 + (100 \times 20) = 3500$

In terms of TR greater than TC, running a pizza stand is profitable for Jaspreet. Estimated profit per week will be Rs.2500/-.

However, this also needs to be compared with benefits foregone before a decision is taken.

Benefits foregone if Jaspreet chose to work for university maintenance crew:

Benefits to Jaspreet for 60 hours of work:

First 40 hours $\times 60 = 2400/-$

Next 20 hours overtime $\times 90 (1.5 \times 60) = 1800/-$

Total benefits foregone: 4200/-.

Therefore, from above calculation it is evident that the opportunity cost for Jaspreet to run pizza stand is 4200/- while the gross benefit comes to 2500 (TR minus TC). If we add opportunity cost into total cost, the net benefits would go in negative. Therefore, Jaspreet should not take pizza stand.

[Total-7]

Solution 38:

The generalized form of budget constraint is as follows:

$$M = P_x X + P_y Y,$$

Where M is income, P_x is price of good X, P_y is price of good Y. Given this, Amit's budget constraint is as follows:

$$3000 = 300B + 400PE$$

Where B stands for book, and PE stands for goods of personal effect. Given this budget constraint, the slope is as follows:

$$\text{Slope of budget constraint} = 300 / 400 = \frac{3}{4}$$

The slope suggest that since the ratio of P_x/P_y is $\frac{3}{4}$, Amit would tend to spend 3000/- in such a manner that he gets benefits (utility) from purchase in the same ratio. The slope also indicates the ratio in which largely the expenditure would be made on books and goods of personal effect.

[Total-5]

Solution 39:

a)

Price elasticity of demand (E_p) = - $[\Delta Q / \Delta P \times P/Q]$

Here,

$P = 50 - 5Q$, hence $\Delta P/\Delta Q = -5$, and $\Delta Q/\Delta P = 1/-5$

Therefore, for $E_p = 1$

$$1 = - [1/-5 \times 50 - 5Q / Q]$$

Solving for Q, we get $Q = 5$, hence Price elasticity of demand will be unitary at output equals to 5.

b)

Given the current sale of 8 units of palm top per period,

$$P = 50 - 5Q = 50 - (5 \times 8) = 10$$

At $P = 10$, price elasticity of demand is as follows:

$$E_p = - [1/-5 \times 10/8] = \frac{1}{4} = 0.25 < 1$$

At the existing price, the demand for palmtop is inelastic and also quite low. For an inelastic product, increase in price would increase the total revenue and hence this is an opportunity for Rashi to increase the price if she wants to increase total revenue.

[Total-8]
